

Pseudobulk Differential Expression Analysis of Intestinal IEL and LPL Compartments: Crohn's Disease versus Control

Bioinformatics Analysis Service 2026-08-05

1. Executive Summary

This report describes the pseudobulk differential expression (DE) analysis performed on the compartment-specific reclustered IEL and LPL single-cell RNA-seq data, comparing Crohn's disease (CD) versus control (Control) tissue within each statistically eligible cell population.

The analytical workflow included:

- aggregation of single-cell counts into per-donor, per-cell-type pseudobulk samples
- a programmatic QC gate requiring at least 2 donors per condition and at least 10 cells per pseudo-sample, blocking any comparison that fails these criteria rather than proceeding silently
- DESeq2 differential expression testing (~ condition design, Control as reference level) with `apeglm` log2 fold-change shrinkage
- diagnostic visualization (volcano plots, `plotMA`, top-DEG heatmaps) and compositional cross-referencing against the upstream cell-type annotation

Six cell populations passed the initial QC gate. Five populations were carried through to DE testing after applying an additional biological interpretability criterion: three in IEL (IEL-Cytotoxic-TRM-like, IEL-TH17, IEL-Cycling-T-cells) and two in LPL (LPL-Cytotoxic-TRM-like, LPL-Treg). Across these five comparisons, differential expression results converge on two recurrent gene-level patterns identified at the compartment-resolved level in this analysis: recurrent up-regulation of *GNLY* in CD across four of five analyzed populations, and a stress-response/glucocorticoid-responsive gene module (*FKBP5*, *TSC22D3*, *TXNIP*, *ZFP36L2*, *NFKBIA*, *DUSP2*) across every cell type analyzed. The latter provides direct, gene-level support for the “response to corticosteroid/glucocorticoid” pathway signal flagged as provisional in the companion GSEA report.

This analysis also surfaces two structural limitations that were not previously quantified: several candidate ambient-RNA/contamination genes among the top differentially expressed genes, and a compartment-specific donor imbalance in which a single CD donor drives the majority of the TH17 signal — a different donor in each compartment. Both are discussed in detail in Section 8.

2. Statistical Design

2.1 Pseudobulk construction and QC gate

For each cell type, single-cell counts were aggregated by donor to form pseudobulk samples. A comparison was only carried forward to DESeq2 testing if it satisfied, for both conditions independently:

- at least 2 donors per condition group
- at least 10 cells per pseudo-sample

This gate is enforced programmatically (`stop()` on failure) rather than as a post-hoc filter, so that no comparison built on an under-powered donor structure is silently reported.

2.2 DESeq2 model

```
design = ~ condition
condition releveled with "Control" as reference
contrast = c("condition", "CD", "Control")
lfcShrink(type = "apeglm")
```

Positive log2 fold change and positive Wald statistic indicate higher expression in CD relative to Control, consistently across all five comparisons (the releveled step is executed independently inside each per-cell-type loop iteration, so the contrast direction cannot be inconsistent between cell types).

2.3 Significance thresholds

Genes were called significant at $\text{padj} < 0.05$ and $|\log_2\text{FoldChange}| > 1$, consistent with the thresholds used to define the significant/DEG gene sets feeding into the companion GSEA analysis.

3. Eligible Cell Types

IEL compartment

Eligible cell types: IEL-Cycling-T-cells, IEL-Cytotoxic-TRM-like, IEL-TH17

LPL compartment

Eligible cell types: LPL-Cytotoxic-TRM-like, LPL-Treg, LPL-Other-CD4-T-cell

LPL-Other-CD4-T-cells passed the minimum QC requirements but was excluded from downstream DE analysis due to its heterogeneous transcriptional composition and lack of a clearly defined biological comparison target.

3.1 Donor structure underlying every comparison

Both compartments draw on the same four donors, split 2 CD / 2 Control:

Donor	Condition	Disease state label
1425	Control	normal
7420	Control	normal
1813	CD	mixed inflamed/non-inflamed (LPL); predominantly non-inflamed (IEL)
1818	CD	mixed inflamed/non-inflamed

Every DESeq2 comparison in this report therefore rests on **exactly the minimum donor count** the QC gate permits (2 vs 2). This is the single most important statistical caveat governing the entire pseudobulk analysis and is revisited in Section 8.1.

Compartment	Cell population	Genes tested	Significant DEGs	Up-regulated	Down-regulated
IEL	IEL-Cycling-T-	4593	1	1	0

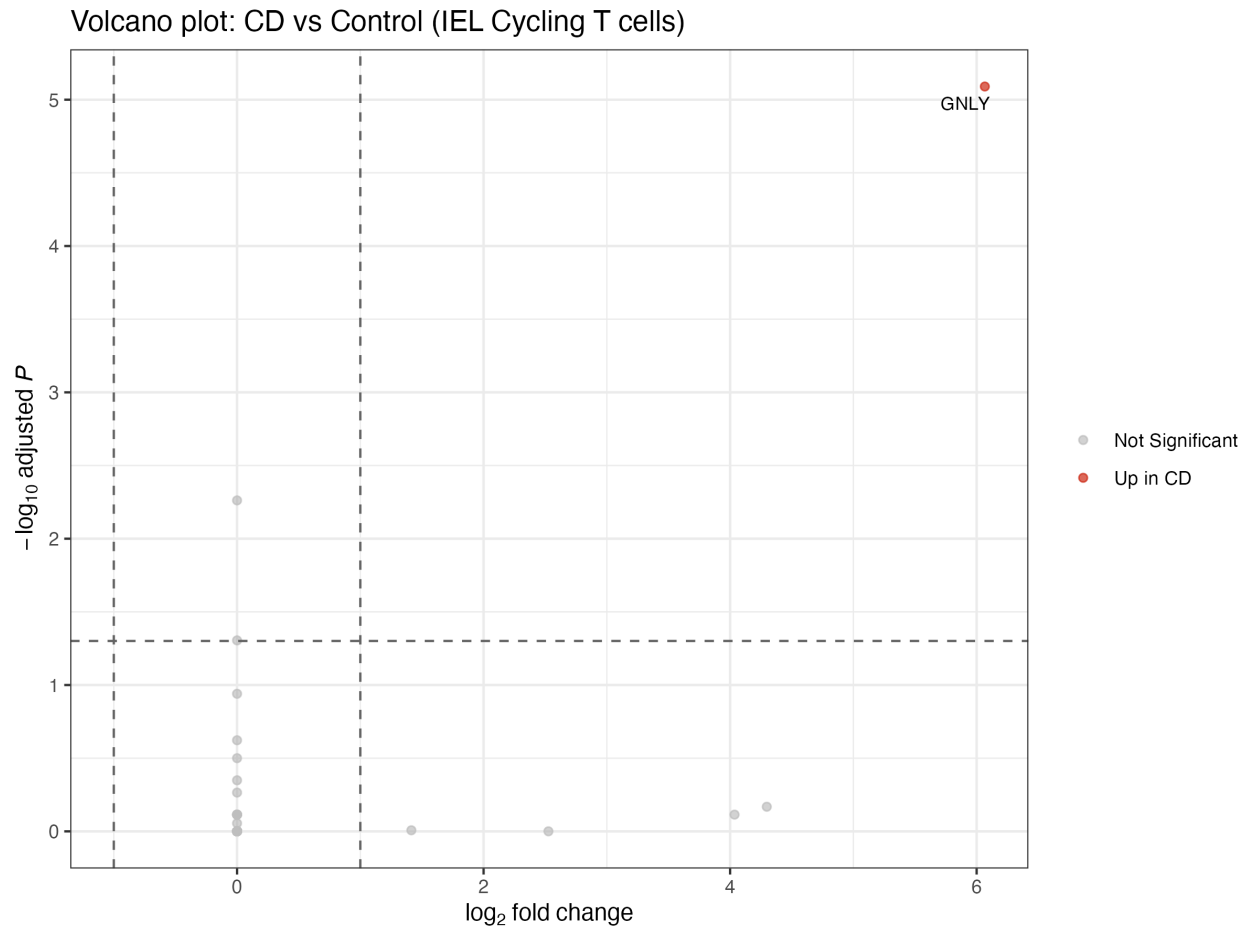
Compartment	Cell population	Genes tested	Significant DEGs	Up-regulated	Down-regulated
	cells				
IEL	IEL-Cytotoxic-TRM-like	11403	10	8	2
IEL	IEL-TH17	10340	5	2	3
LPL	LPL-Cytotoxic-TRM-like	11413	14	6	8
LPL	LPL-Treg	9432	2	0	2

Table 1. Summary of the differential expression analysis performed for each eligible cell population. *Genes tested* indicates the number of genes retained after pre-filtering and included in the DESeq2 statistical model. Significant DEGs were defined as genes with an adjusted *P* value < 0.05 and an absolute log2 fold change > 1.

4. IEL Compartment: Differential Expression Results

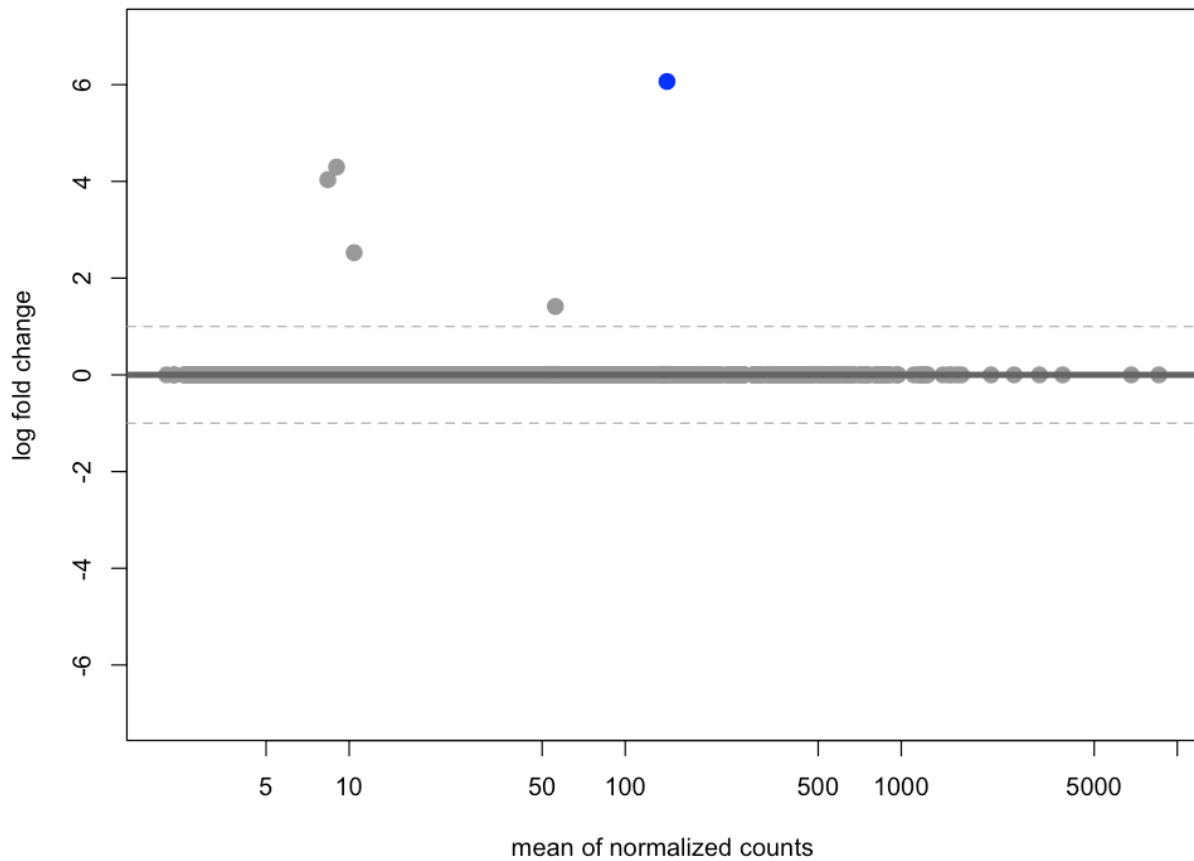
4.1 IEL-Cycling-T-cells

Volcano plot

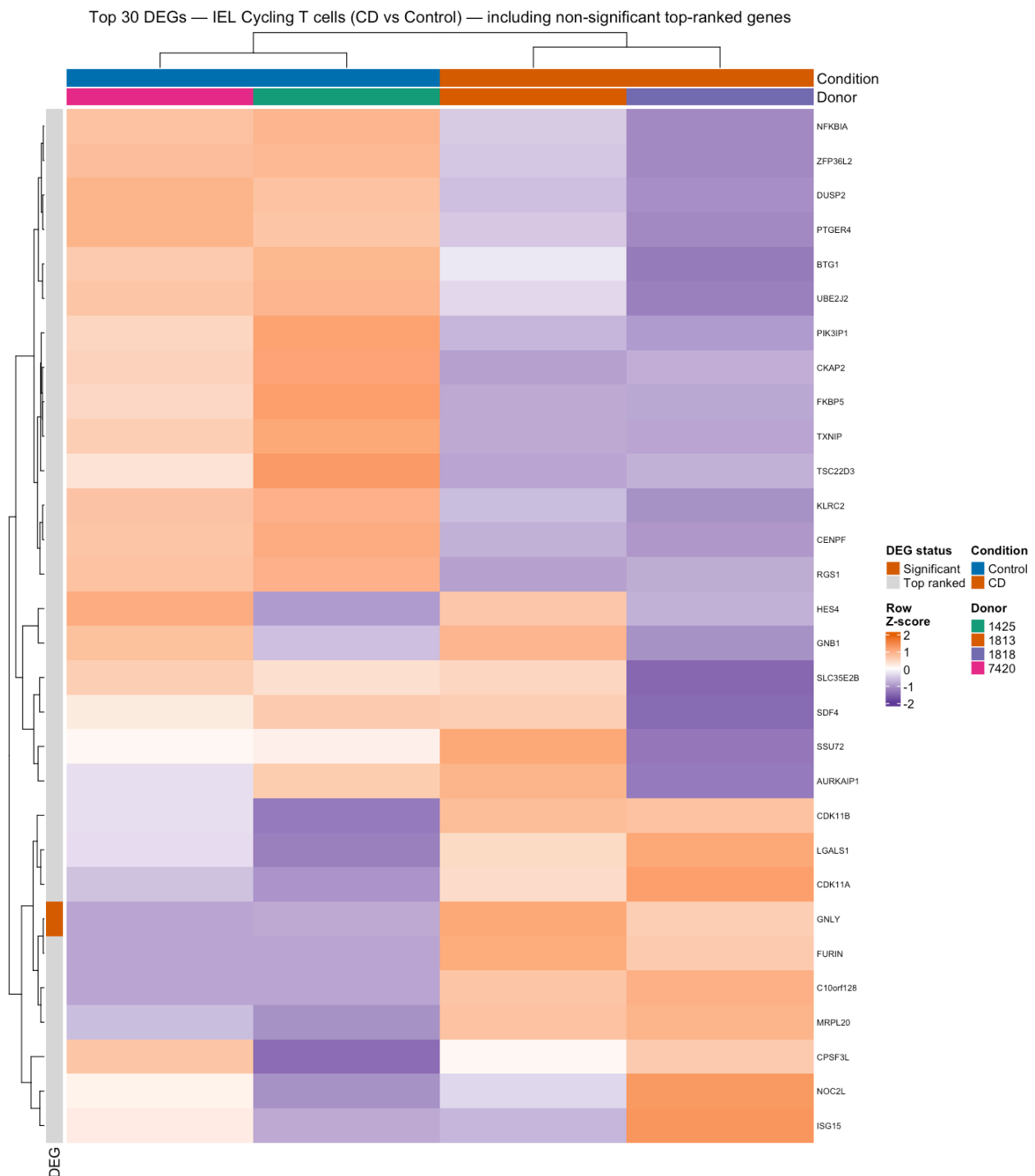


MA plot (shrunk LFC)

MA plot: CD vs Control (IEL Cycling T cells) — shrunk LFC



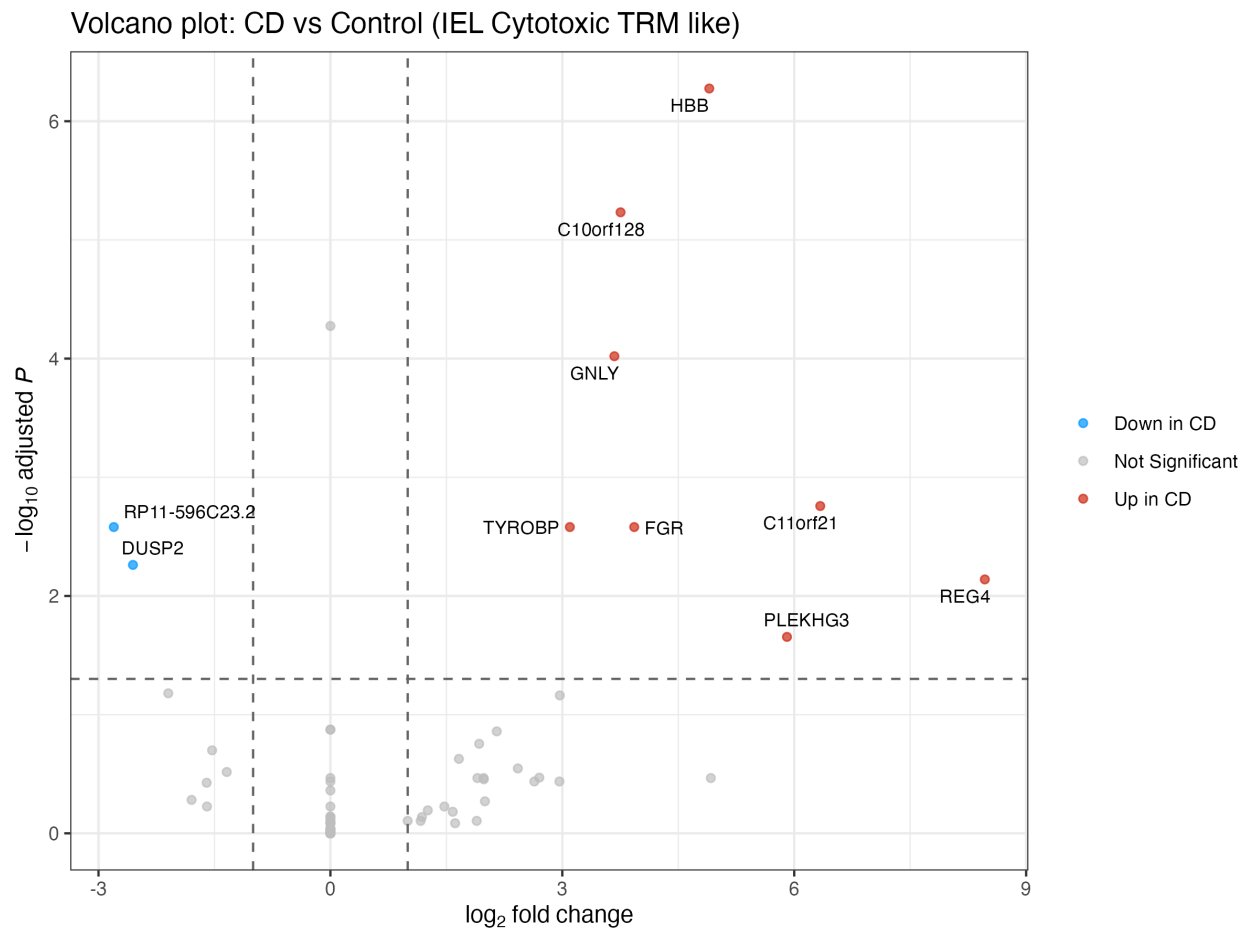
Top 30 DEGs (including non-significant top-ranked genes)



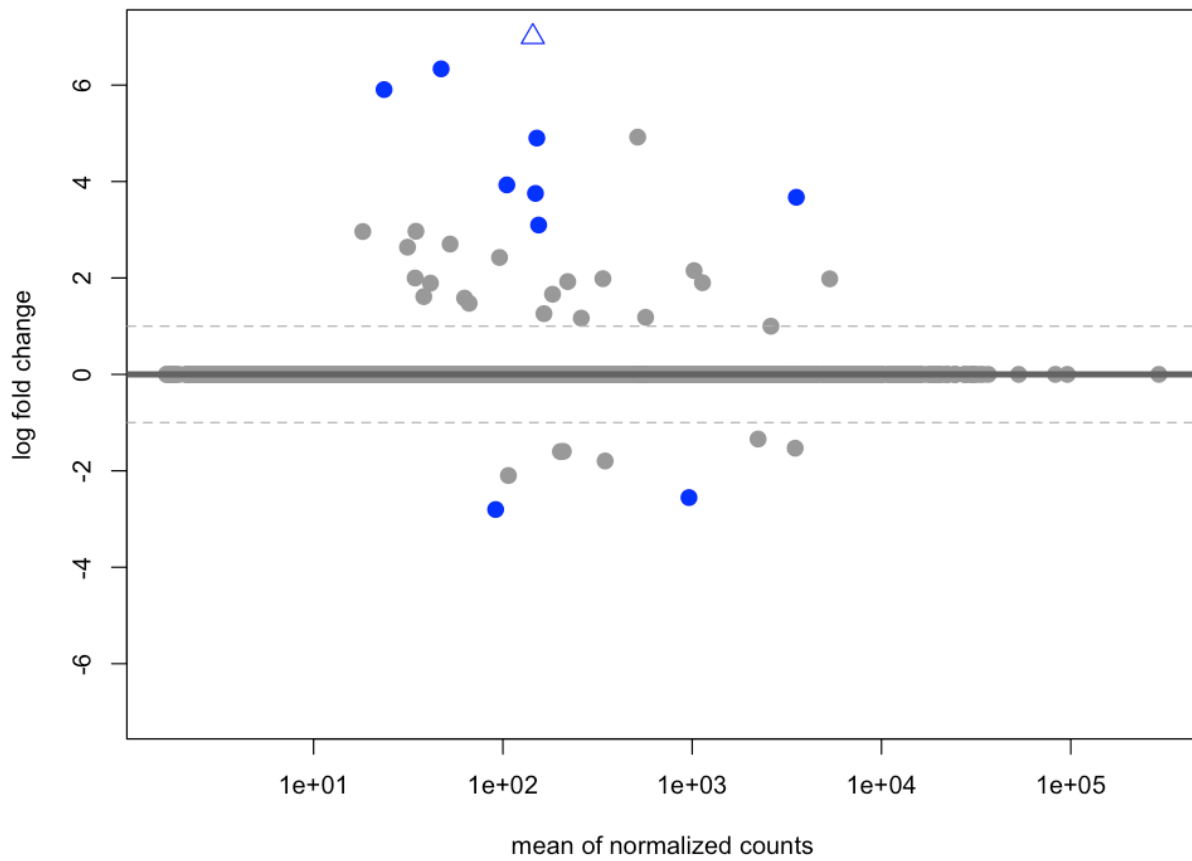
Interpretation. Only one gene reaches significance: *GNLY* (granulysin), strongly up in CD. This is consistent with the small leading-edge sizes observed for this population in the companion GSEA analysis, and with this being the smallest of the five eligible cell types by cell number. The `plotMA` shows `apeglm` shrinkage behaving as expected: the bulk of genes are pulled tightly around LFC = 0, with only a handful of low-to-moderate count genes retaining appreciable fold change. The top-30 heatmap shows the glucocorticoid-responsive module (*NFKBIA*, *ZFP36L2*, *DUSP2*, *PTGER4*, *FKBP5*, *TXNIP*, *TSC22D3*) consistently higher in Control than CD across both donors per condition, despite none of these genes individually crossing the significance threshold here — a pattern-level signal below individual-gene significance, consistent with why GSEA (Section 4.3 of the companion report) detects this axis even where single-gene DE testing does not.

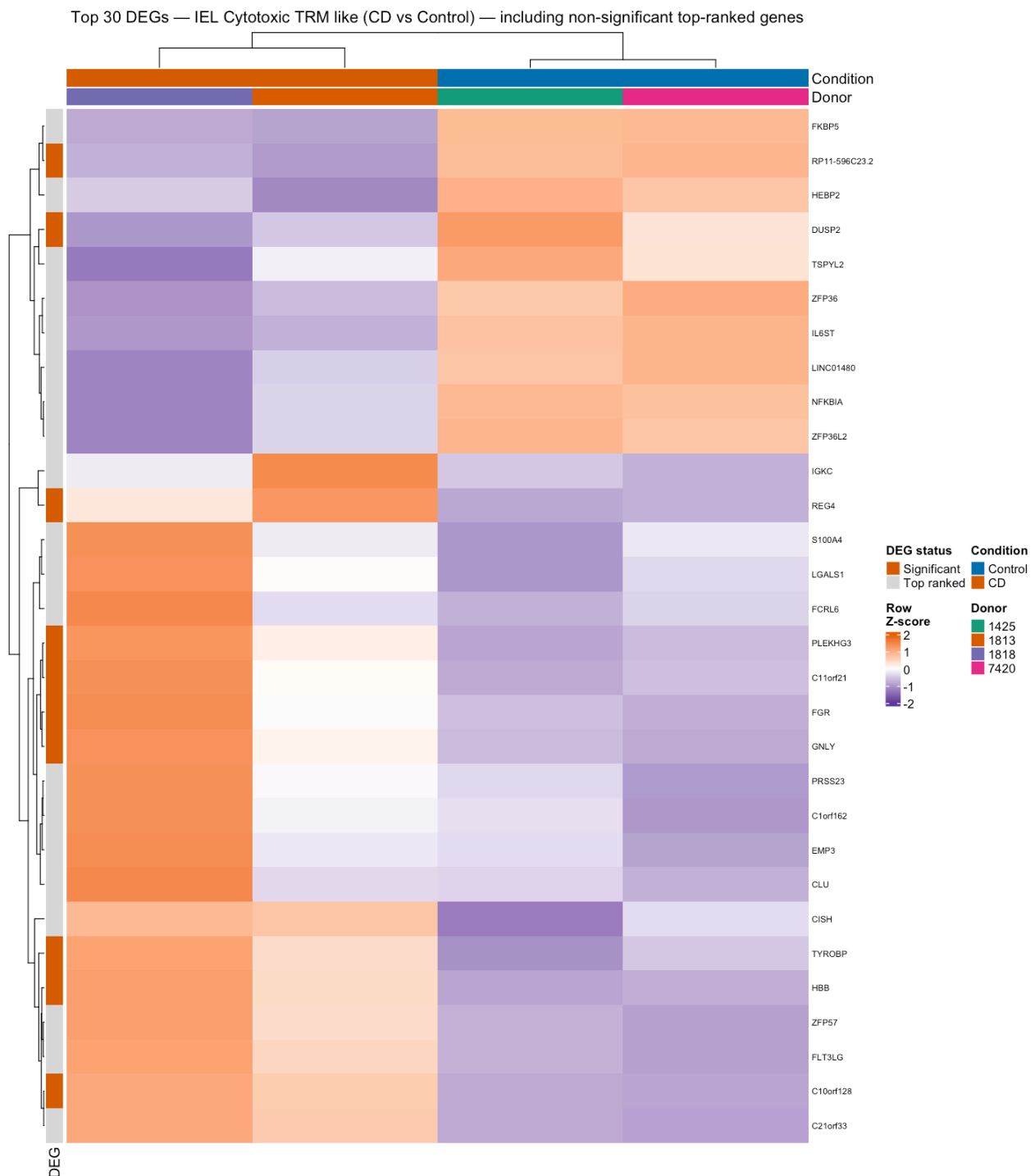
4.2 IEL-Cytotoxic-TRM-like

Volcano plot



MA plot (shrunk LFC)

MA plot: CD vs Control (IEL Cytotoxic TRM like) — shrunk LFC**Top 30 DEGs (including non-significant top-ranked genes)**



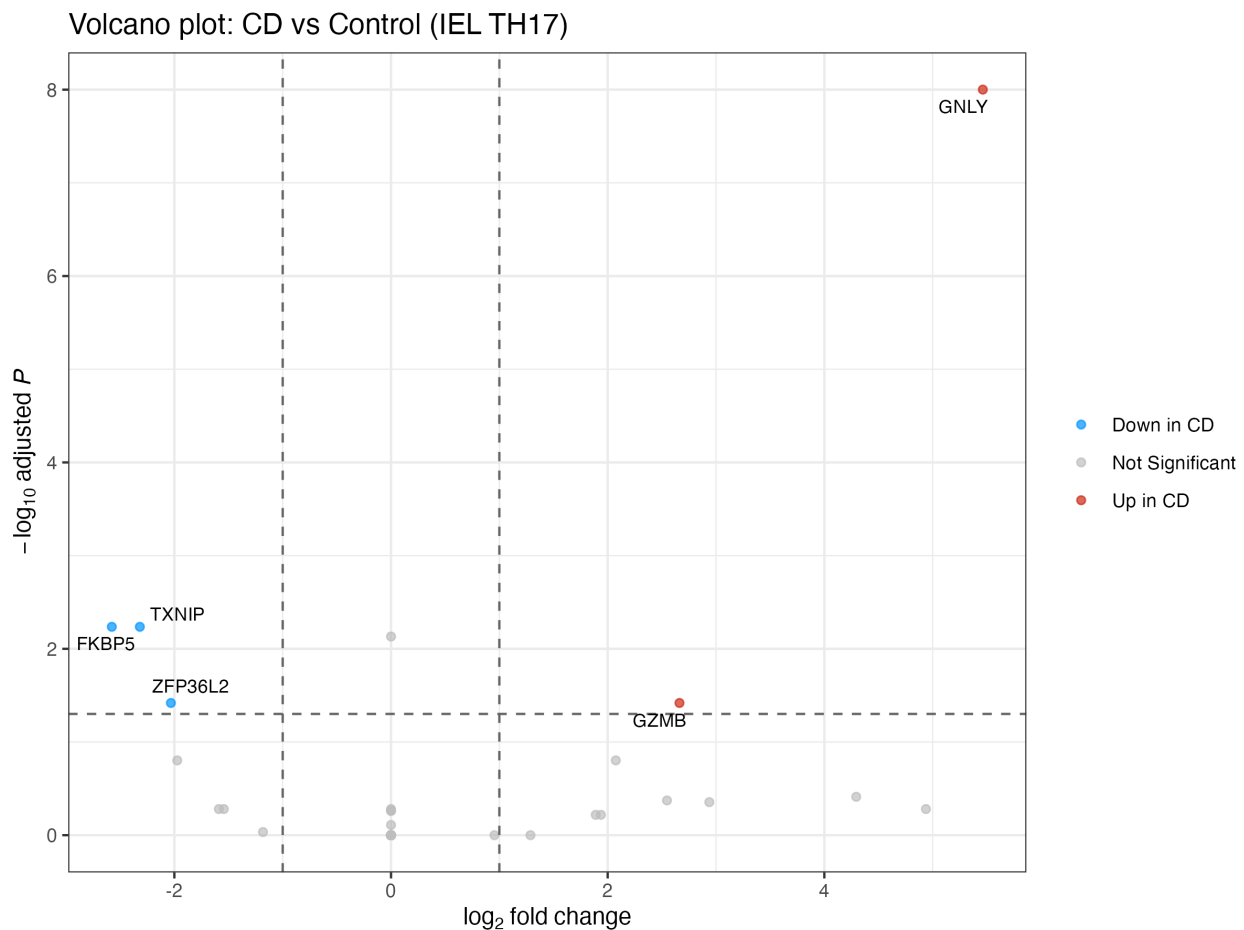
Interpretation. This comparison shows the second-largest number of significant DEGs among the five (visually: 8 up — *HBB*, *C10orf128*, *GNLY*, *TYROBP*, *FGR*, *C11orf21*, *REG4*, *PLEKHG3* — and 2 down — *RP11-596C23.2*, *DUSP2*). Two findings require explicit caution rather than direct biological interpretation:

- ***HBB*** (hemoglobin beta) among the top up-regulated genes is a recognized signature of ambient erythrocyte RNA contamination in droplet-based scRNA-seq, not a plausible T-cell transcriptional program.
- ***REG4*** (an epithelial antimicrobial peptide gene) is biologically implausible as a genuine T-cell transcript and is more consistent with ambient contamination from adjacent epithelial cells — physically expected given the intraepithelial localization of this compartment.

TYROBP and *FGR* (myeloid signaling adaptors) up in CD are more ambiguous: they could reflect genuine expression in NK-like tissue-resident cytotoxic cells (documented in some cytotoxic T/NK-like populations) or residual ambient signal; this cannot be resolved from pseudobulk data alone (Section 8.2). The top-30 heatmap again shows the corticosteroid-responsive module (*FKBP5*, *NFKBIA*, *ZFP36L2*, *ZFP36*, *IL6ST*) elevated in Control relative to CD.

4.3 IEL-TH17

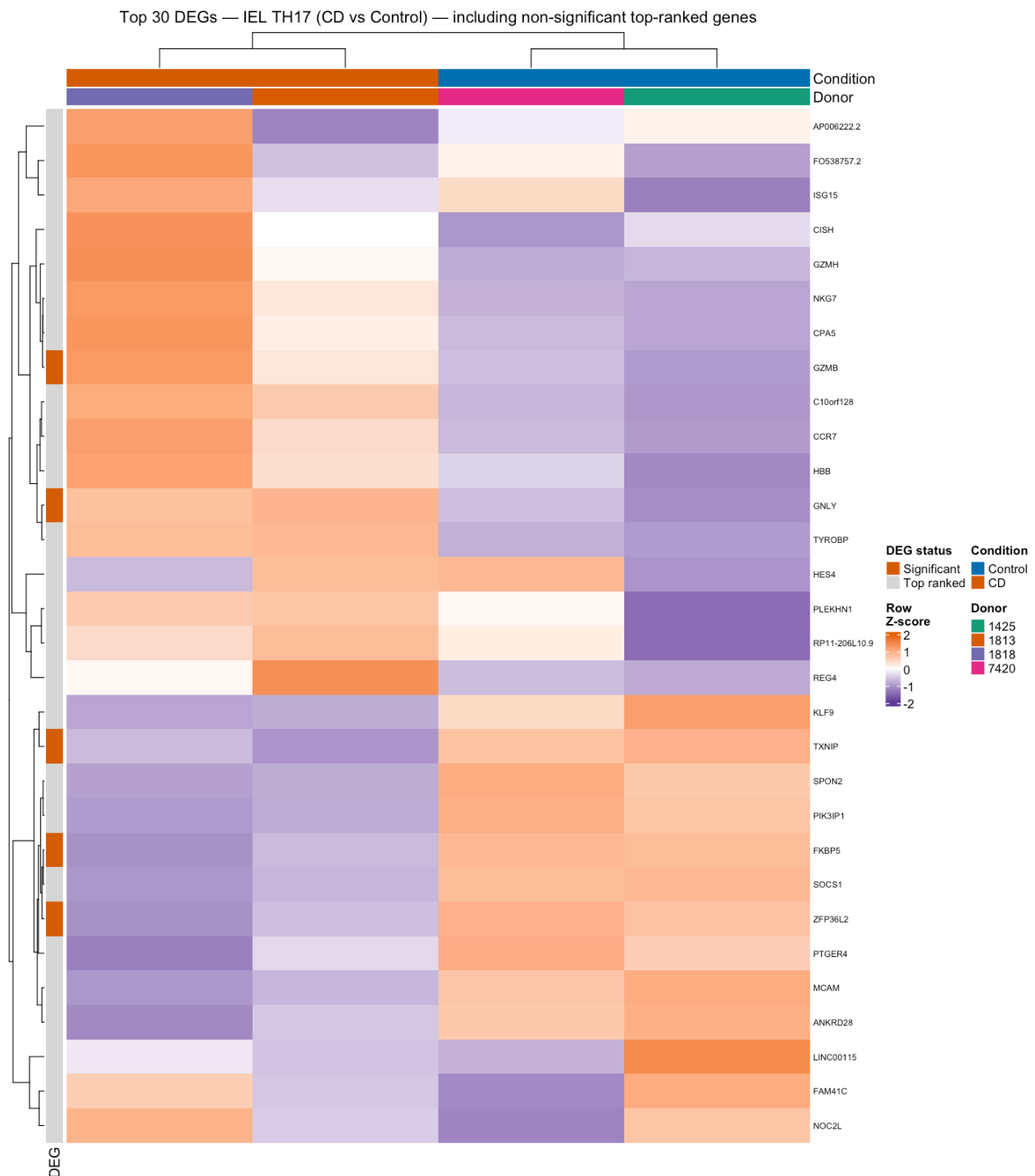
Volcano plot



MA plot (shrunk LFC)

A scatter plot showing the relationship between the mean of normalized counts (x-axis, logarithmic scale from 1e+01 to 1e+05) and the log fold change (y-axis, linear scale from -6 to 6). The plot displays 100 data points, with 10 points highlighted in blue. A solid horizontal line is drawn at y=0, and two dashed horizontal lines are drawn at y=1 and y=-1. The blue points are scattered across the plot, with some showing high log fold change values (e.g., around 5.5) and others showing low values (e.g., around -2.5).

file:///Users/davidemaccarrone/Desktop/Bioinformatics/MyProjects👉/scRNA-Seq/UPDATE/reports/Pseudobulk_analysis/REPORT_Pseudobulk.html

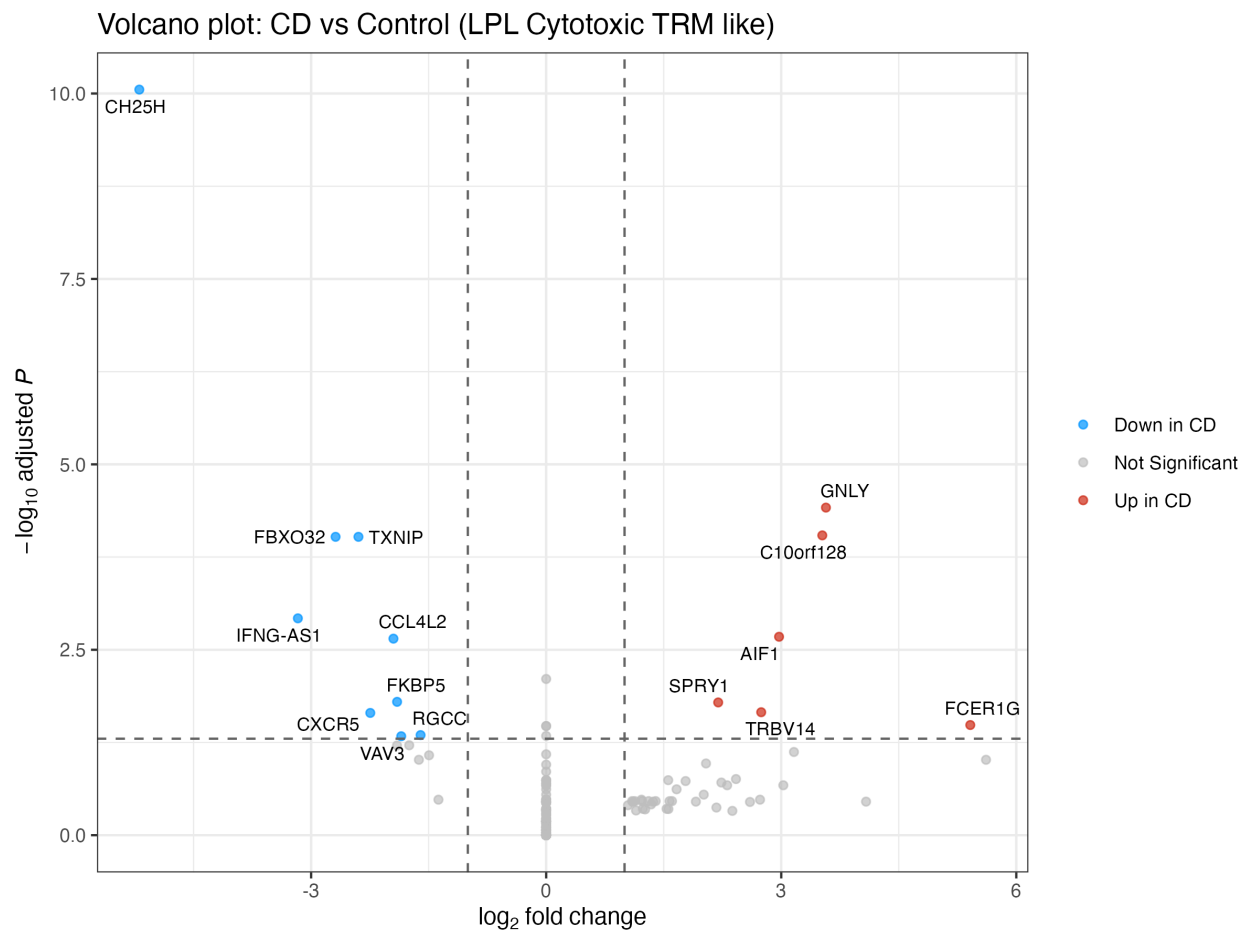


Interpretation. Five significant genes (visually: *GNLY*, *GZMB* up in CD; *FKBP5*, *TXNIP*, *ZFP36L2* down in CD). *GZMB* up is consistent with the activated, cytotoxic-leaning TH17 phenotype described in the companion GSEA report and in Jaeger et al. (2021). The three down-regulated genes are all members of the glucocorticoid-responsive module discussed above, again individually significant here (unlike in IEL-Cycling-T-cells), reinforcing that this module's signal is strong enough in this population to clear single-gene significance, not only pathway-level significance. **This comparison should be interpreted with the donor-imbalance caveat in Section 8.3:** cell-type composition analysis indicates the CD arm of this comparison is dominated (~67% of TH17 cells) by a single donor (1818).

5. LPL Compartment: Differential Expression Results

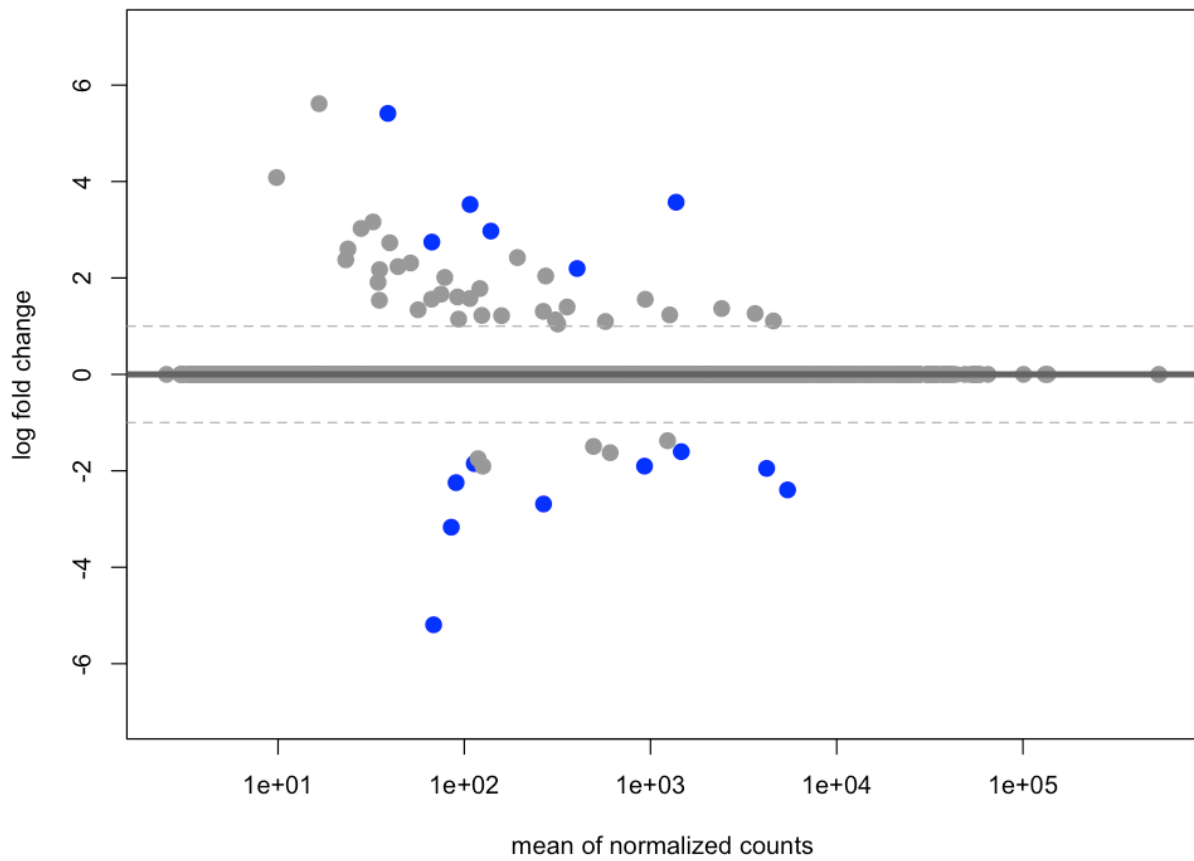
5.1 LPL-Cytotoxic-TRM-like

Volcano plot

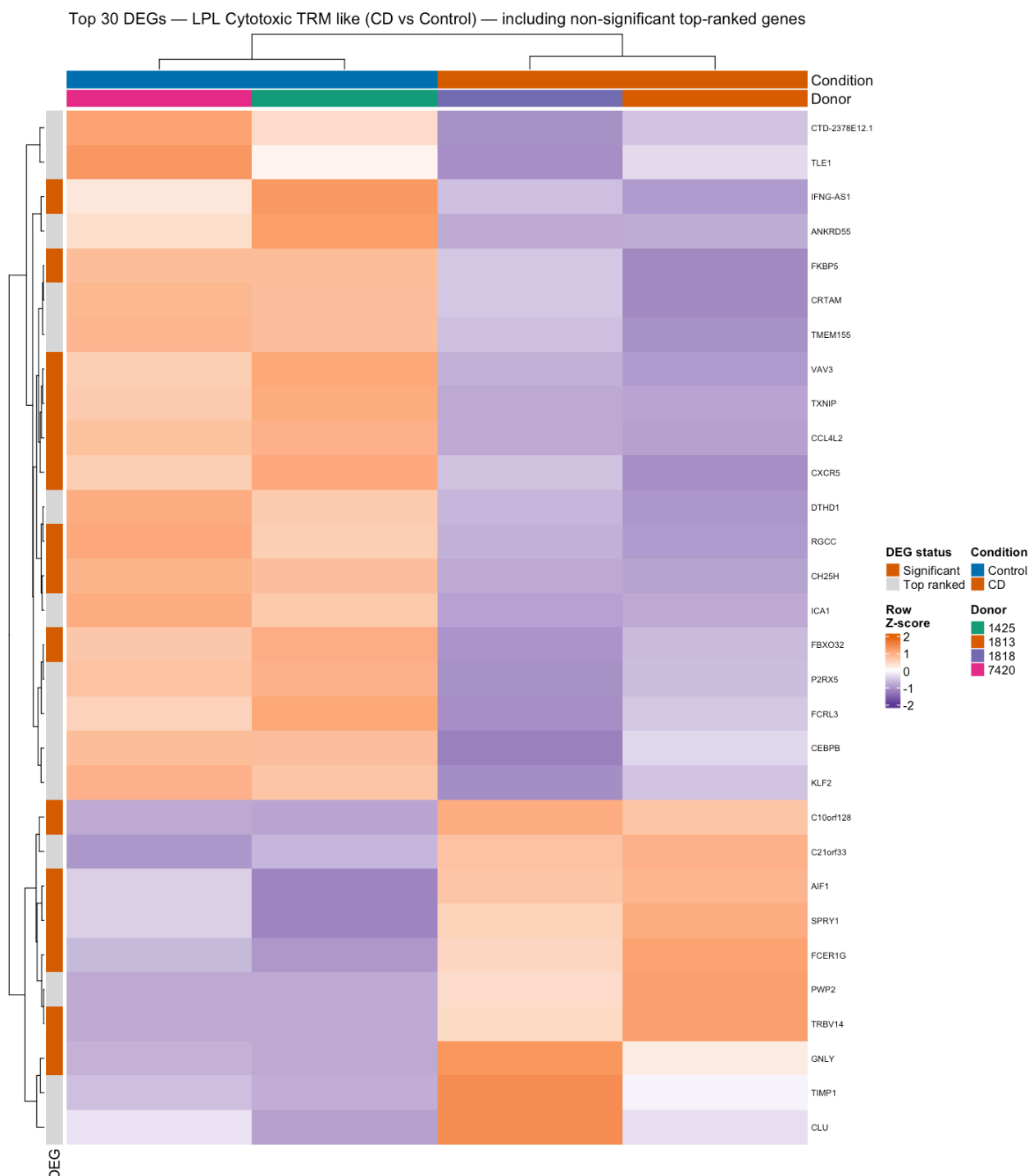


MA plot (shrunk LFC)

MA plot: CD vs Control (LPL Cytotoxic TRM like) — shrunk LFC



Top 30 DEGs (including non-significant top-ranked genes)



Interpretation. This is the comparison with the largest number of significant DEGs (visually: 6 up — *GNLY*, *C10orf128*, *AIF1*, *SPRY1*, *TRBV14*, *FCER1G*; 8 down — *CH25H*, *FBXO32*, *TXNIP*, *IFNG-AS1*, *CCL4L2*, *FKBP5*, *CXCR5*, *RGCC*), consistent with this being one of the larger eligible populations. Two individual results warrant a specific note rather than being read at face value alongside the pathway-level IFN- γ up-regulation reported in the GSEA analysis of this same population (Section 5.1, companion report):

- ***IFNG-AS1*** (a long non-coding RNA reported to positively regulate *IFNG* transcription) is significantly *down* in CD, which sits in apparent tension with the Hallmark Interferon Gamma Response signature being *up* in CD for this same population. These are not necessarily contradictory (the lncRNA is not itself a member of the Hallmark gene set, and negative-feedback dynamics are plausible),

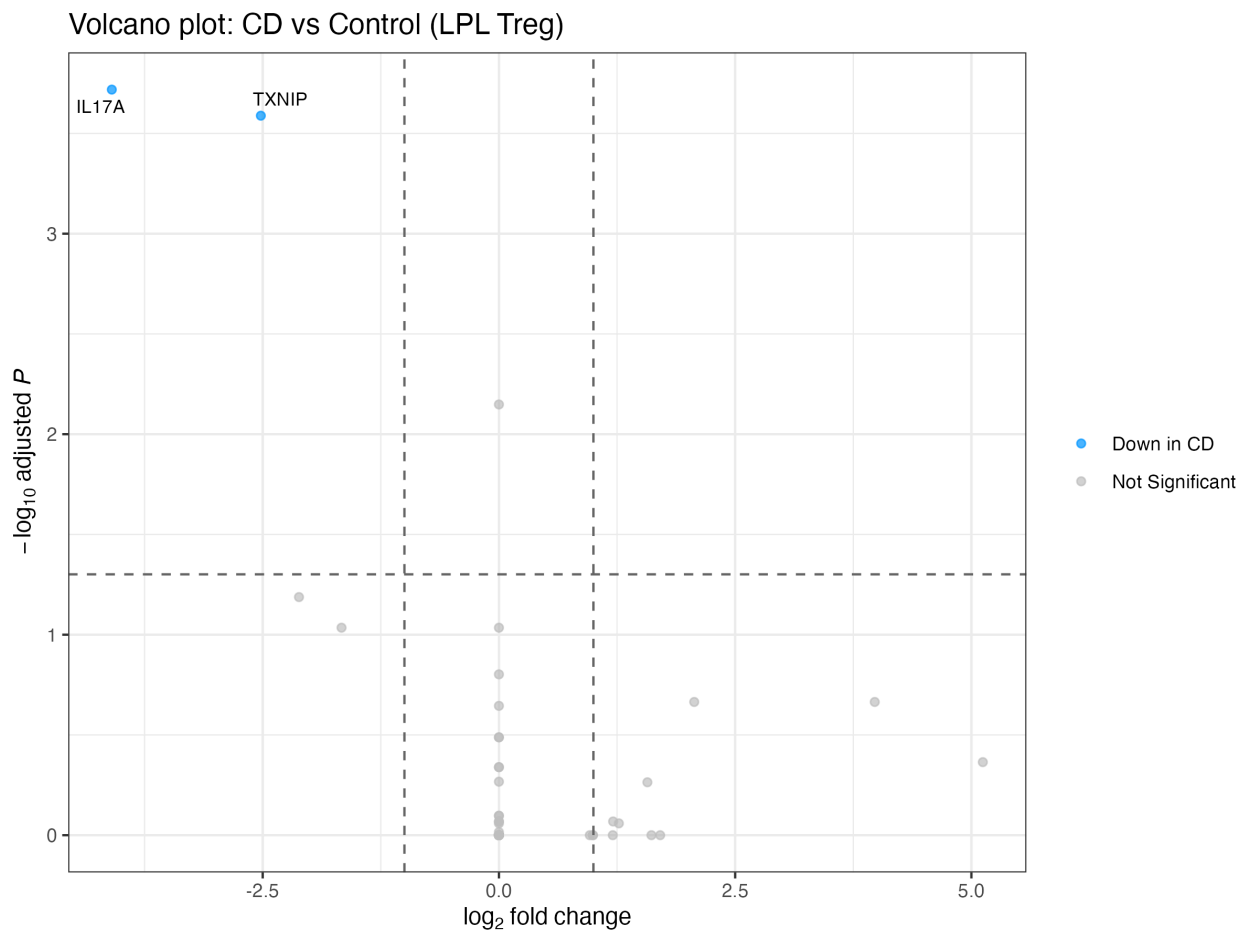
but the tension should be resolved by inspecting IFN- γ pathway leading-edge genes directly rather than assumed away.

- **CH25H** (an interferon-stimulated gene with immunomodulatory, oxysterol-mediated function) is the single most significant gene in this entire report and is down in CD — again notable alongside an up-regulated IFN- γ response signature, and worth the same leading-edge-level reconciliation.

The corticosteroid-responsive module (*FKBP5*, *TXNIP*) recurs here as well, consistent with every other comparison in this report.

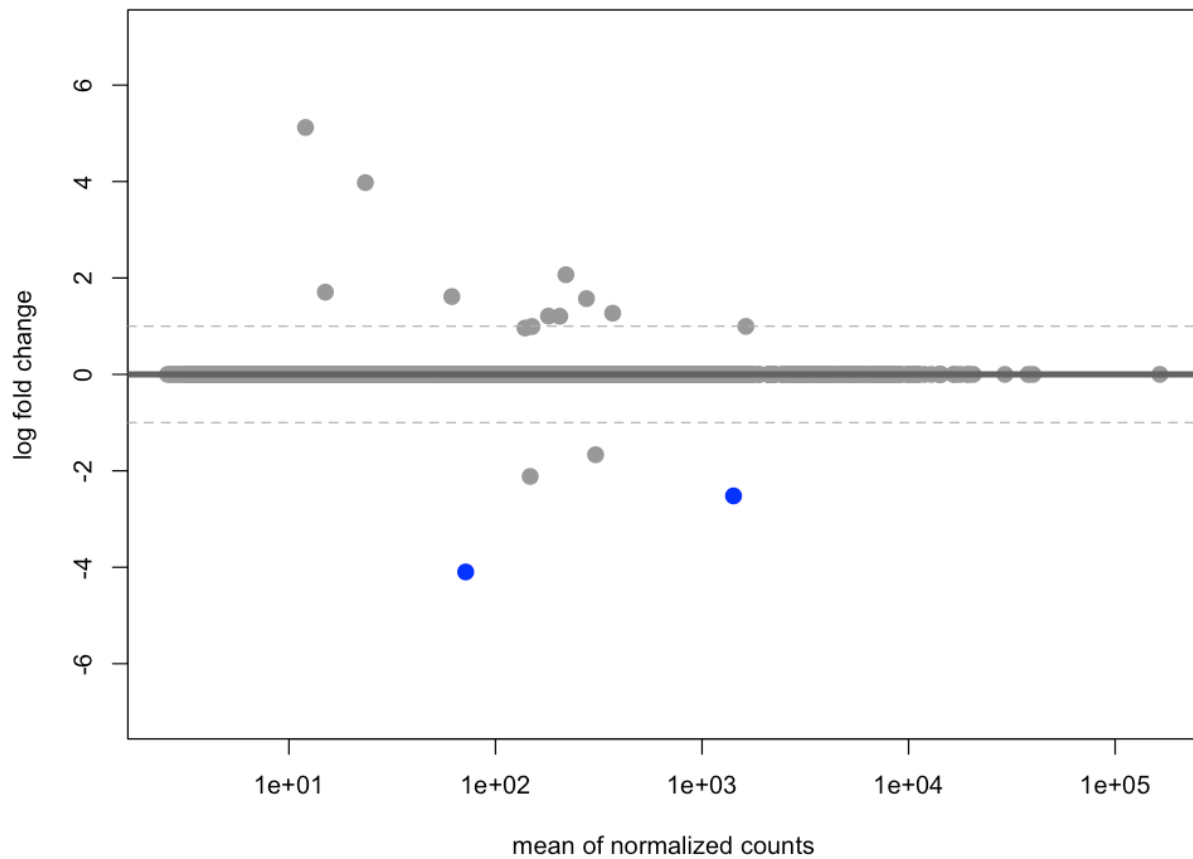
5.2 LPL-Treg

Volcano plot



MA plot (shrunk LFC)

MA plot: CD vs Control (LPL Treg) — shrunk LFC



Top 30 DEGs (including non-significant top-ranked genes)

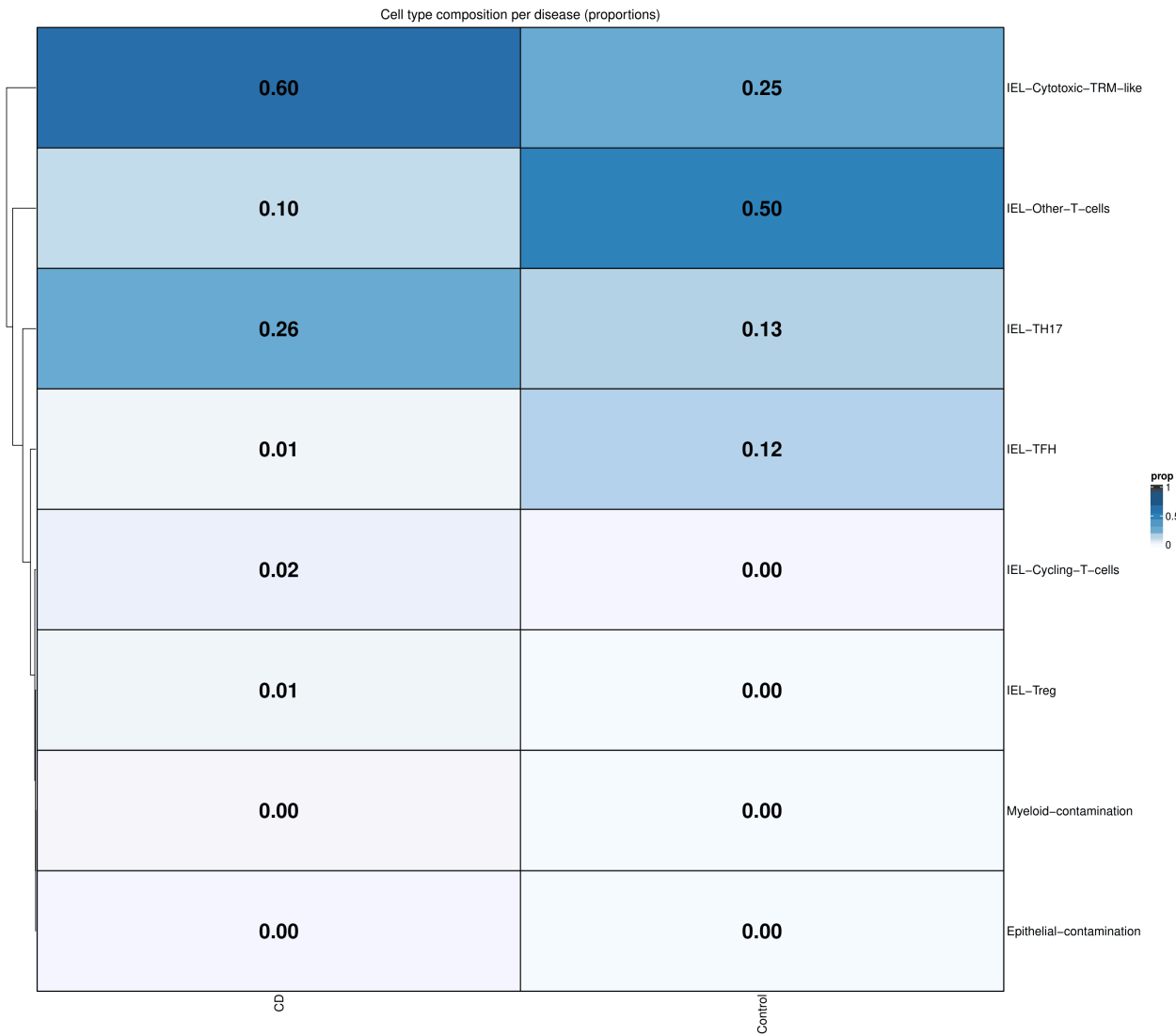


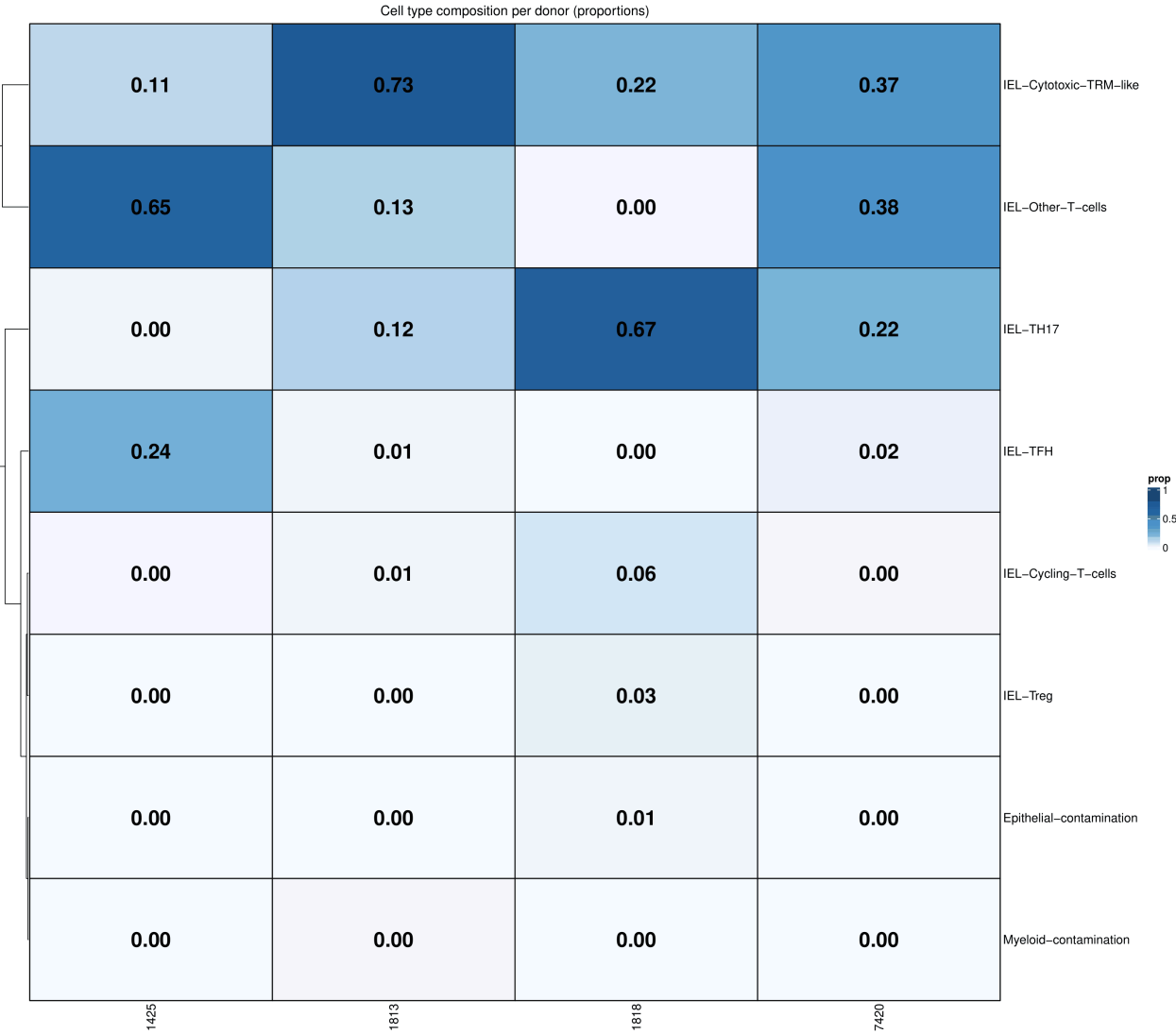
Interpretation. The fewest significant genes of any comparison in this report: only *IL17A* and *TXNIP*, both down in CD, with no genes significantly up. *TXNIP* down is consistent with the glucocorticoid-responsive module pattern seen throughout. *IL17A* down in a Treg-annotated population is worth flagging rather than over-interpreting: it could reflect a reduction in IL17A-producing Treg/Th17-plasticity cells (a population with precedent in the literature, e.g. LAG3⁺ Tr1 cells producing regulatory cytokines alongside residual type-17 markers) within CD, consistent in direction with the reduced pathway-level TH17/Treg balance regulation implied by the IL2-STAT5 signaling finding in the companion GSEA report — but this is a single gene in a population with very few total significant hits, and should be treated as a lead for follow-up rather than a settled finding.

6. Compositional Context

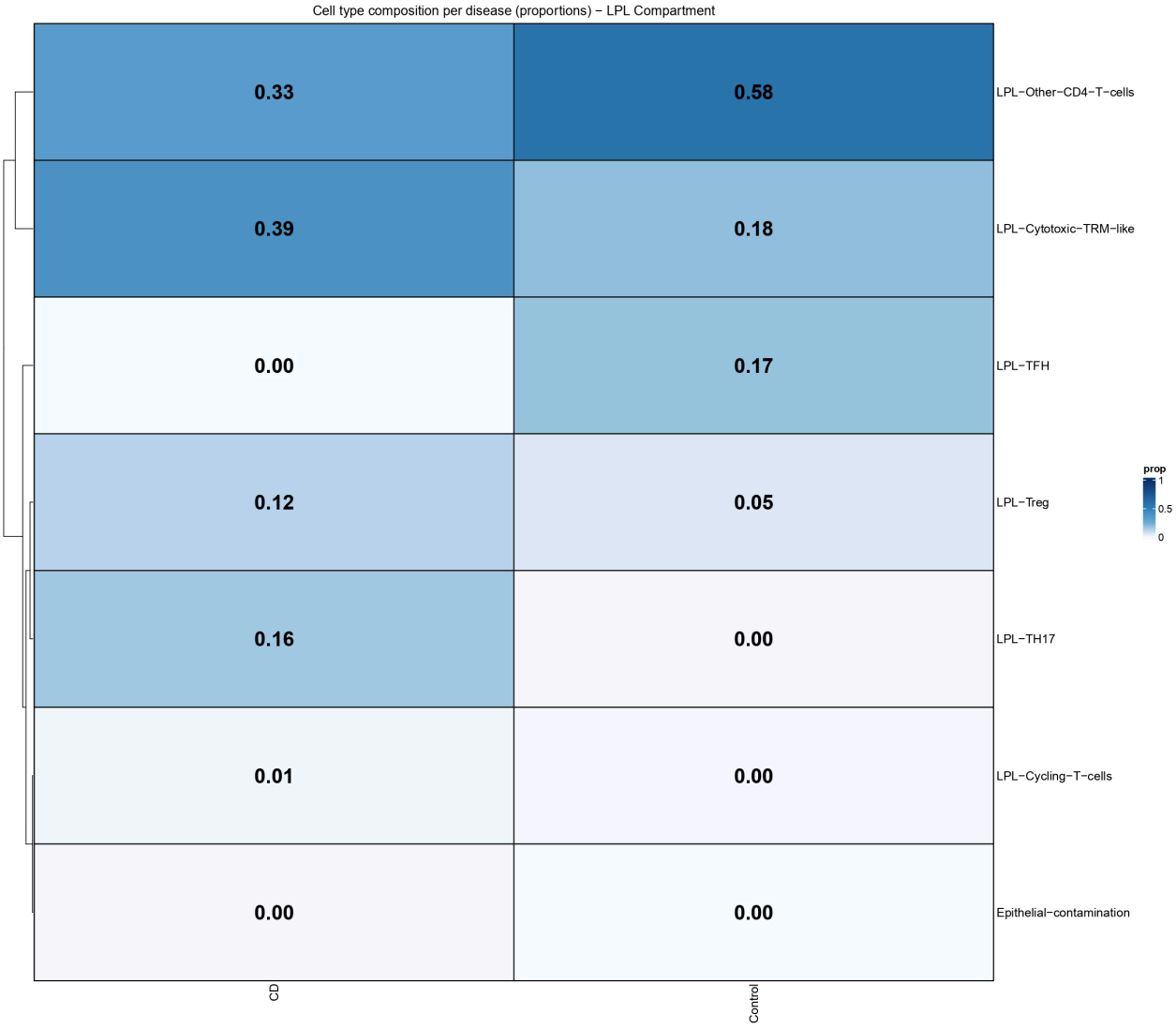
6.1 Compartment composition by condition

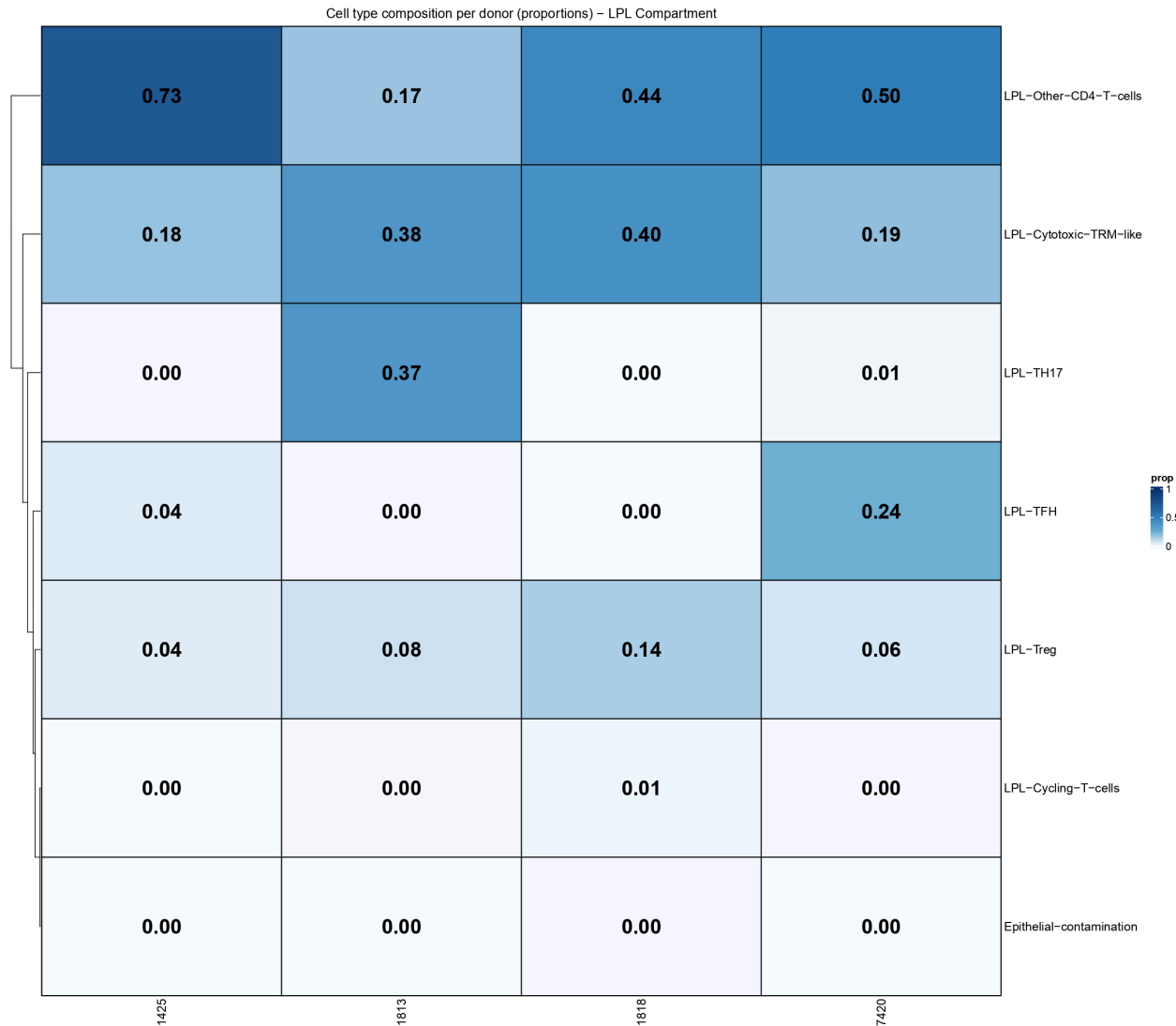
IEL — proportions by disease and condition/donor





LPL — proportions by disease and donor





6.2 Cross-compartment compositional concordances

Two compositional shifts are directionally concordant across both compartments and independently support the DE/GSEA findings above:

Cell type	IEL: CD vs Control	LPL: CD vs Control	Direction
TFH	1% vs 12%	0% vs 17%	Concordant loss in CD, both compartments
Cytotoxic-TRM-like	60% vs 25%	39% vs 18%	Concordant relative expansion in CD, both compartments

One compositional shift is **not** concordant with the a priori expectation set by the companion GSEA report’s citation of reduced Treg proportion in CD (Jaeger et al. 2021):

Cell type	IEL: CD vs Control	LPL: CD vs Control
Treg	1% vs 0%	12% vs 5%

Both compartments show a *higher*, not lower, Treg proportion in CD by this pipeline’s transcriptional clustering. This is flagged explicitly as an open discrepancy in Section 8.4 rather than reconciled by assumption.

7. Comparison with GSEA Results

Pseudobulk DE finding	Corresponding GSEA finding	Relationship
Glucocorticoid-responsive module (FKBP5, TSC22D3, TXNIP, ZFP36L2, NFKBIA, DUSP2) down in CD, all 5 cell types	“Response to corticosteroid/glucocorticoid” down, 3/5 cell types (GO:BP)	Direct gene-level confirmation; resolves the “not independently verified” caveat (Section 10.5, GSEA report)
GNLY up in CD, 4/5 cell types	Cytotoxic/granzyme-associated GO:BP and Hallmark terms up across IEL/LPL cytotoxic populations	Consistent individual driver gene
GZMB up in IEL-TH17	Hallmark Inflammatory Response / IFN-γ Response up, IEL-TH17	Consistent individual driver gene
IFNG-AS1, CH25H down in CD, LPL-Cytotoxic-TRM-like	Hallmark Interferon Gamma Response up, same population	Apparent tension; requires leading-edge reconciliation (Section 5.1)
TXNIP down in CD, LPL-Treg	IL2-STAT5 Signaling up (Hallmark), LPL-Treg	Compatible (TXNIP is a negative regulator of glucose uptake/STAT signaling contexts)

Pseudobulk DE finding	Corresponding GSEA finding	Relationship
		but not directly tested

8. Limitations

8.1 Minimum-donor statistical power

Limitations of the pseudobulk analysis

Every DESeq2 comparison in this report is based on **2 CD and 2 Control donors**. Although **at least three biological replicates per condition are generally recommended** for robust RNA-seq differential expression analysis, only two donors per group were available after quality-control filtering and were therefore retained for the analysis. DESeq2 remains computationally applicable with two replicates, although dispersion estimation and inference are less reliable.

This represents the principal limitation of the pseudobulk differential expression analysis. With $n = 2$ biological replicates per condition, DESeq2 has limited statistical power to estimate biological variability accurately and to distinguish true disease-associated effects from donor-specific variation. Accordingly, the small number of significant DEGs identified in each comparison (typically 1–14 genes) is expected and should not be interpreted as evidence for the absence of biological differences.

For this reason, **gene set enrichment analysis (GSEA)** constitutes the primary biological readout of this study. Because GSEA evaluates the complete ranked gene list rather than relying exclusively on individual gene significance thresholds, it can provide a more informative view of coordinated biological changes in low-powered settings such as this. Consequently, the pseudobulk differential expression results should be interpreted primarily as **gene-level support for pathway-level alterations identified by GSEA**, rather than as an independent discovery layer with equivalent statistical weight.

8.2 Candidate ambient-RNA contamination genes

HBB and *REG4* among the top DEGs in IEL-Cytotoxic-TRM-like are better explained by ambient erythrocyte and epithelial RNA contamination, respectively, than by genuine T-cell biology. *TYROBP*, *FGR*, *FCER1G*, and *AIF1* (myeloid signaling adaptors) recurring across cytotoxic populations are more ambiguous and cannot be resolved as genuine versus ambient from pseudobulk data alone. Recommended follow-up: cross-check these genes' expression specificity against the myeloid-contamination cluster already identified during upstream QC (Subsetting Compartment report), and consider explicit exclusion or flagging of established ambient-RNA marker genes before final reporting.

8.3 Compartment-specific donor imbalance in TH17

Cross-referencing the compositional heatmaps with the DE results reveals that the TH17 comparison is driven by a different single CD donor in each compartment: donor 1818 contributes ~67% of IEL-TH17 cells (versus ~12% from donor 1813), while donor 1813 contributes ~37% of LPL-TH17 cells (versus near-zero from donor 1818). With only 2 CD donors total, this means each TH17 comparison is closer to a single-donor signal with a second, sparsely-represented donor, than to a balanced 2-donor comparison. This should be stated explicitly alongside any TH17-specific finding in this or the companion GSEA report, and is a stronger, more specific version of the general minimum-donor caveat in Section 8.1.

8.4 Treg compositional direction discrepancy

As detailed in Section 6.2, this dataset shows Treg proportion *increasing*, not decreasing, in CD in both compartments — in tension with the Jaeger et al. (2021) flow cytometry finding of reduced CD25^{hi} CD4⁺ T cells in CD, cited as supporting context in the companion GSEA report's interpretation of the LPL-Treg IL2-STAT5 signal (Section 5.2, GSEA report). A plausible reconciliation is that this pipeline's transcriptionally-defined Treg cluster and the original paper's CD25^{hi} flow cytometry gate are not measuring identical populations, but this has not been verified and should be treated as an open discrepancy, not assumed away.

9. Reproducibility

Environment:

- R version: 4.6.0
- DESeq2 version: 1.52.0
- apegglm version: 1.34.0

All intermediate DE result objects are stored under:

objects/pseudobulk/

All visual outputs under:

IEL Compartment: results/plots/Pseudobulk_IEL/

LPL Compartment: results/plots/Pseudobulk_LPL/

10. Deliverables

- Pseudobulk DESeq2 result tables (`res` , `res_shrunk`) per cell type, per compartment
- Volcano plots, MA plots, top-ranked gene expression heatmaps per cell type (this report)
- Compositional proportion tables (cell type $\times$ condition / disease / donor) for both compartments

11. Conclusion

Pseudobulk differential expression analysis across five cell populations in the IEL and LPL compartments identifies a small, statistically constrained but biologically coherent set of CD-associated genes. Two individual-gene-level patterns — near-universal *GNLY* up-regulation and a coordinated glucocorticoid-responsive module down-regulation — recur across four of five analyzed populations and directly corroborate pathway-level findings from the companion GSEA analysis, most notably resolving the previously-flagged “not independently verified” status of the corticosteroid response signature. At the same time, this analysis surfaces concrete, previously undocumented limitations — a compartment-specific single-donor dominance of the TH17 signal, several candidate ambient-RNA genes among top DEGs, and a Treg compositional trend running opposite to the cited literature — that should accompany any downstream use of these results and are proposed as priority follow-up items (Section 8).